AI-Based Emotion Recognition from Text

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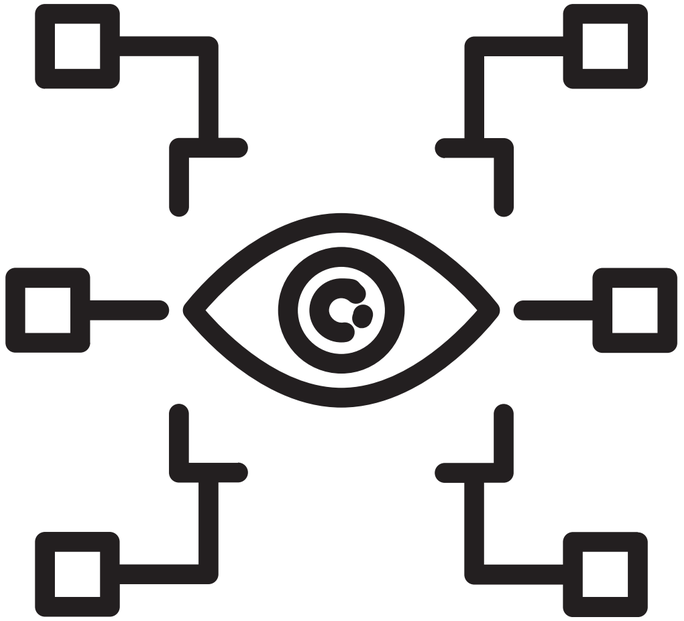
AI-Batch 8

Problem Statement

- Text contains hidden emotional meaning
- Manual emotion detection is inefficient
- Social media produces massive textual data
- Need automated emotion classification system

Project Objective

- Build a machine learning model to classify emotions
- Use labeled dataset from Kaggle
- Convert text into numerical features
- Deploy an interactive web application



Dataset Overview

- Source: Kaggle Emotion Dataset
- 16,000 training samples
- 2,000 validation samples
- 2,000 testing samples
- 6 emotion classes:
- Sadness
- Joy
- Love
- Anger
- Fear
- Surprise

	text	label
0	i didnt feel humiliated	0
1	i can go from feeling so hopeless to so damned...	0
2	im grabbing a minute to post i feel greedy wrong	3
3	i am ever feeling nostalgic about the fireplac...	2
4	i am feeling grouchy	3

```
label
1    5362
0    4666
3    2159
4    1937
2    1304
5     572
Name: count, dtype: int64
```

Model Architecture

User Text Input (Raw sentence entered by user) → TF-IDF Vectorization (Convert text → numerical features) → Logistic Regression Model (Classifies emotion) → Emotion + Confidence Score (Final output to user)

- TF-IDF converts text into numeric features
- Logistic Regression classifies emotion
- Output includes predicted emotion + confidence score

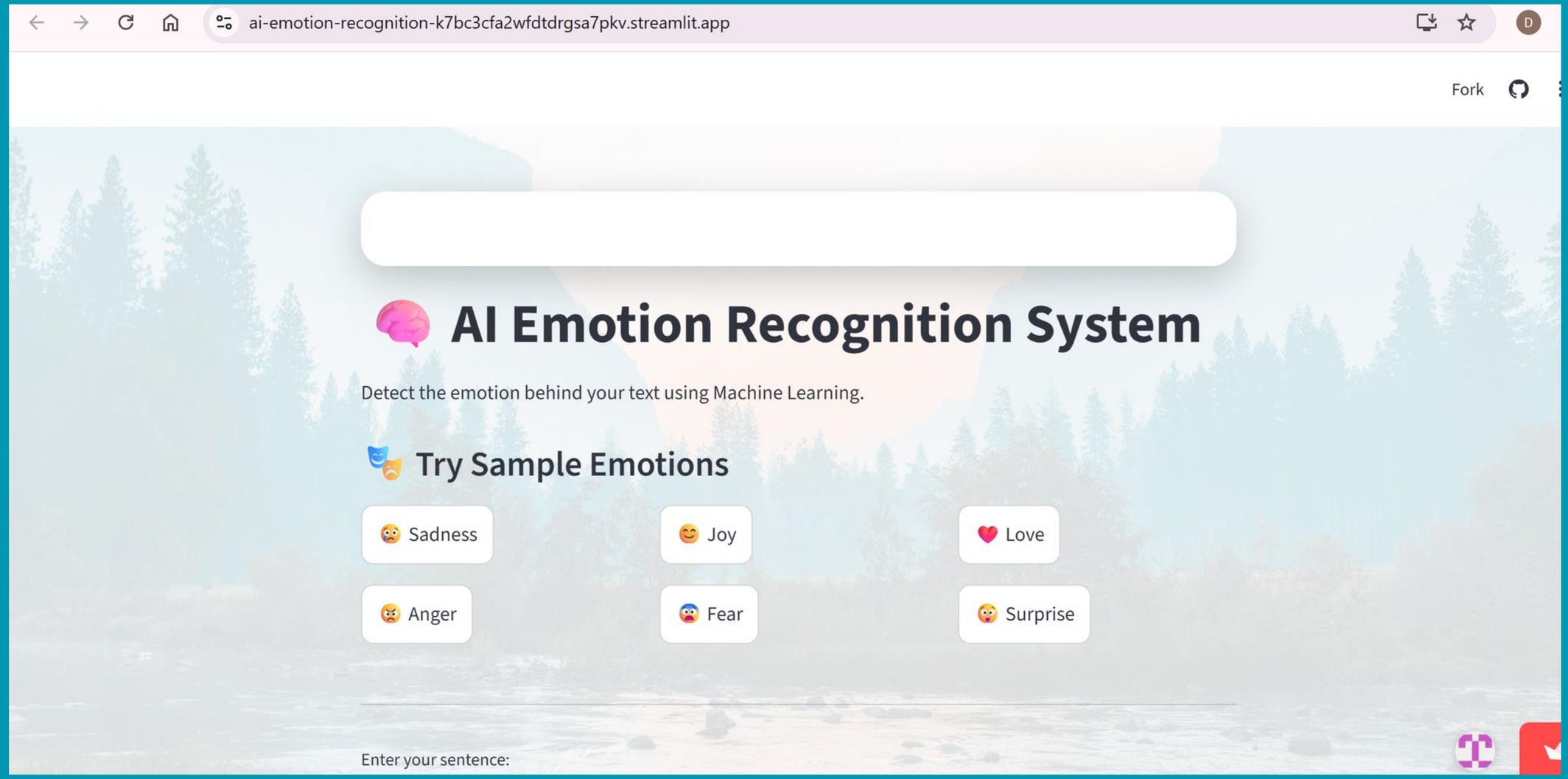
Model Evaluation

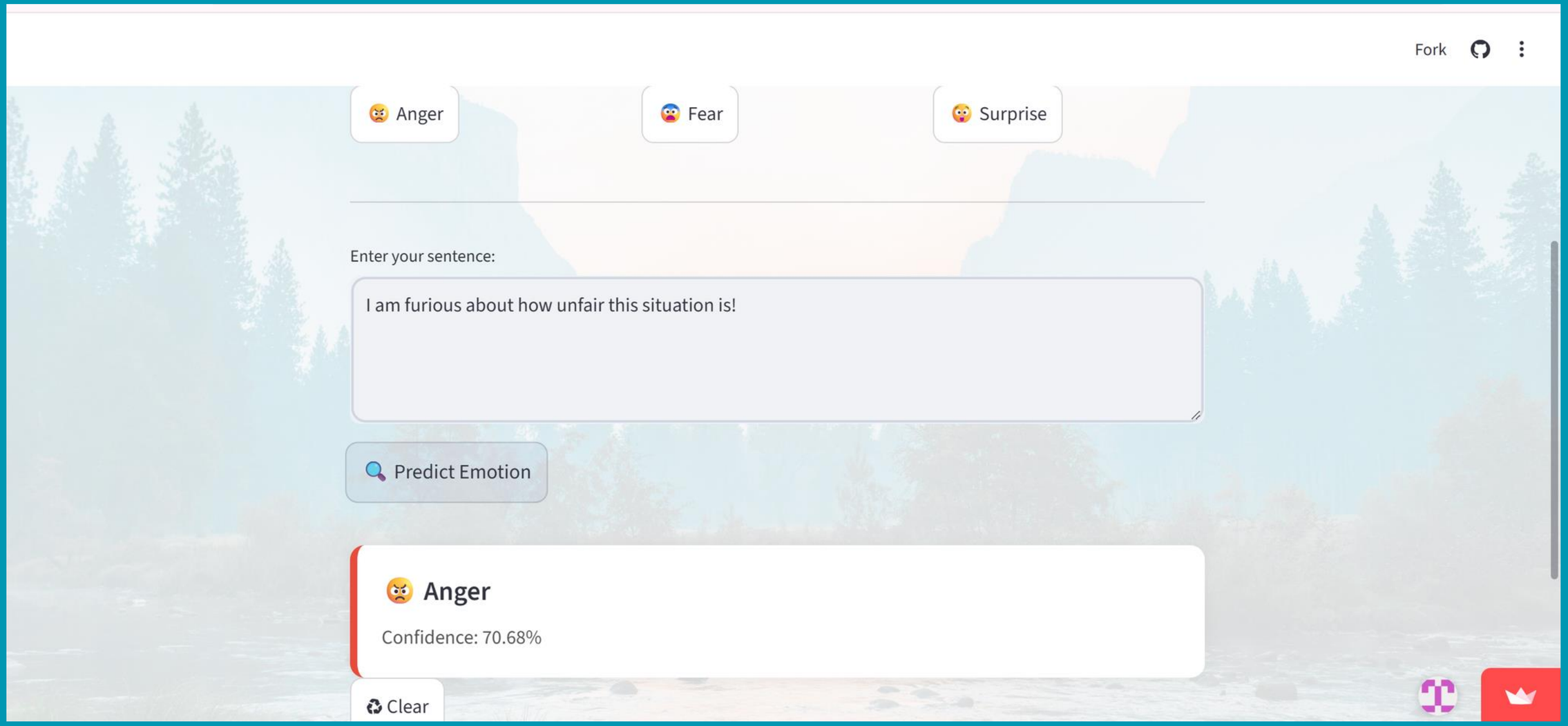
- Accuracy: 89%
- Evaluated on unseen test data
- Strong performance on majority classes
- Used classification report for evaluation

```
Test Accuracy: 89.2 %

Classification Report:

              precision    recall  f1-score   support

           0       0.92      0.94      0.93       581
           1       0.87      0.97      0.92       695
           2       0.84      0.67      0.74       159
           3       0.90      0.84      0.87       275
           4       0.91      0.85      0.88       224
           5       0.97      0.52      0.67        66

    accuracy                           0.89      2000
   macro avg       0.90      0.80      0.84      2000
weighted avg       0.89      0.89      0.89      2000


Model saved successfully.
```

Deployment

- Built interactive UI using Streamlit
- User can enter custom text
- Shows predicted emotion with confidence
- Deployed on Streamlit Community Cloud


ai-emotion-recognition-k7bc3cfa2wfdtdrgsa7pkv.streamlit.app
Fork
AI Emotion Recognition System
Detect the emotion behind your text using Machine Learning.
Try Sample Emotions
Sadness
Joy
Love
Anger
Fear
Surprise
Enter your sentence:


Fork
Anger
Fear
Surprise
Enter your sentence:
I am furious about how unfair this situation is!
Predict Emotion
Anger
Confidence: 70.68%
Clear

Conclusion

- Successfully built end-to-end ML pipeline
- Applied text preprocessing and vectorization
- Trained and evaluated classification model
- Deployed real-world AI application

Thank You